Quiz-3 | CSE715 28/08/2026

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(a) $X = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$, $P = \begin{bmatrix} 0.1 & 0.0 \\ 0.0 & 0.1 \\ 0.1 & 0.1 \end{bmatrix}$, $\tilde{X} = X + P$

$$\therefore \tilde{X} = X + P = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} + \begin{bmatrix} 0.1 & 0.0 \\ 0.0 & 0.1 \\ 0.1 & 0.1 \end{bmatrix} = \begin{bmatrix} 1.1 & 0.0 \\ 0.0 & 1.1 \\ 1.1 & 1.1 \end{bmatrix}$$

(b) since, $\omega_Q = \omega_K = I_2$, we have

$Q = \tilde{X}$

$K = \tilde{X}$

$V = \tilde{X}$

$q_2 = [0.0, 1.1]$ query vector position for second row

unscaled dot product $q_2 K^T$:

here, $K^T = \begin{bmatrix} 1.1 & 0.0 & 1.1 \\ 0.0 & 1.1 & 1.1 \end{bmatrix}$

$$q_2 K^T = \begin{bmatrix} 0.0, 1.1 \end{bmatrix} \begin{bmatrix} 1.1 & 0.0 & 1.1 \\ 0.0 & 1.1 & 1.1 \end{bmatrix}$$

$$= \begin{bmatrix} (0.0 \times 1.1) + (1.1 \times 0.0), (0.0 \times 0.0) + (1.1 \times 1.1), (0.0 \times 1.1) \\ + (1.1 \times 1.1) \end{bmatrix}$$

$$\therefore q_2 K^T = [0.0, 1.21, 1.21]$$

Scale by $\sqrt{d_k}$ - $d_k = 2$ $\therefore$ $\sqrt{d_k} = \sqrt{2} = 1.4142$

$\therefore S_{21} = \frac{0.6}{1.4142} = 0.0000$

$S_{22} = \frac{1.21}{1.4142} = 0.8556$

$S_{23} = \frac{1.21}{1.4142} \approx 0.8556$

$S_2 = [0.000, 0.8556, 0.8556]$

(c) Here,

$M_2 = [0, 0, -\infty]$

add M_2 to S_2 :

$S_2 + M_2 = [0.0 + 0, 0.8556 + 0, 0.8556, -\infty]$

Applying the Softmax function :

$e^0 = 1.0$, $e^{0.8556} \approx 2.3528$, $e^{-\infty} = 0.0$

Sum = 1.0 + 2.3528 + 0.0 = 3.3528

probabilities :

$\alpha_{2,1} = \frac{1.00}{3.3528} \approx 2.983$

$\alpha_{2,2} = \frac{2.3528}{3.3528} \approx 0.7017$

$\alpha_{2,3} = \frac{0.0}{3.3528} \approx 0.0000$

$\therefore \alpha_2 \approx [0.2983, 0.7017, 0.0000]$

(d) Self Attention Output

$$z_2 = \alpha_2 V$$

$$V = \tilde{x} = \begin{bmatrix} 1.1 & 0.0 \\ 0.0 & 1.1 \\ 1.1 & 1.1 \end{bmatrix}$$

$$z_2 = [0.2983, 0.7017, 0.0000] \times \begin{bmatrix} 1.1 & 0.0 \\ 0.0 & 1.1 \\ 1.1 & 1.1 \end{bmatrix}$$

$$\Rightarrow z_2 = 0.2983 \times [1.1, 0.0] + 0.7017 \times [0.0, 1.1] + 0 \times [1.1, 1.1]$$

$$\Rightarrow z_2 = [0.3281, 0.0000] + [0.0000, 0.7719] + [0.0000, 0.0000]$$

$$\Rightarrow z_2 \approx [0.3281, 0.7719]$$ (Ans)